Blood vessels of thorax (descending thoracic aorta & internal thoracic artery azygous system of veins)

By

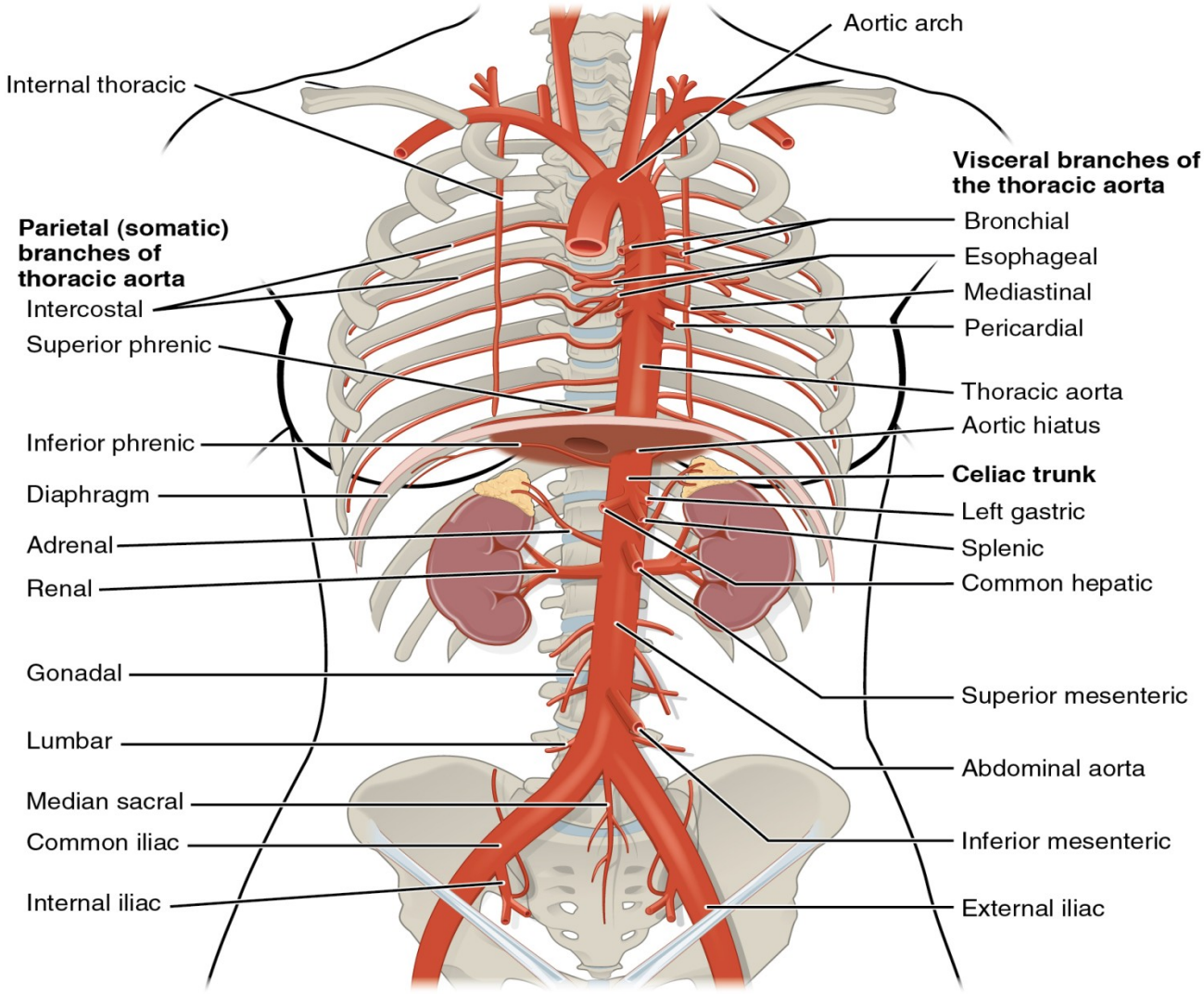
Dr. MOHNI BAI

OBJECTIVES

- At the end of the class students should able to understand

1. Origin, course , termination & branches of the descending thoracic aorta.
2. Origin , course, termination & branches of the internal thoracic artery.
3. Origin, course, termination & tributaries of the azygous & hemiazygous system of the veins.
4. Some clinical correlations.

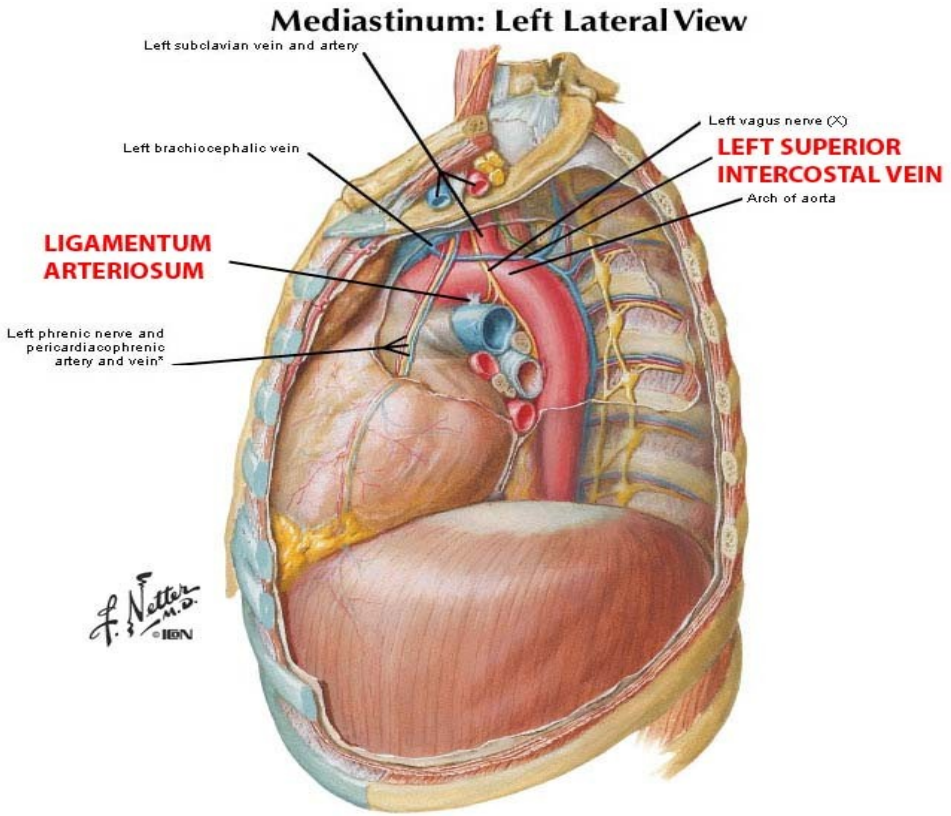
Descending thoracic aorta



Descending thoracic aorta

- Origin.

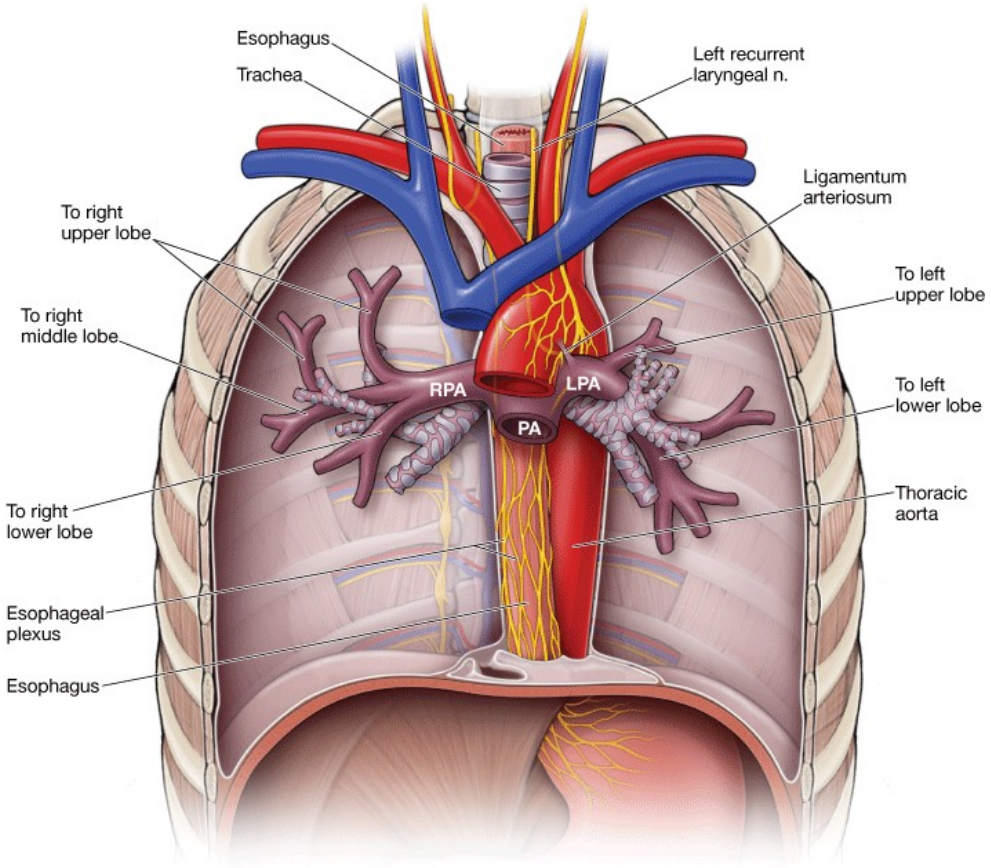
1. Is continuation of the arch of the aorta.
2. Begins on the left side of the lower border of the body of the fourth thoracic vertebra.
3. Lies in the posterior mediastinum.
4. Descend with inclination at right.



Descending thoracic aorta

- Termination.

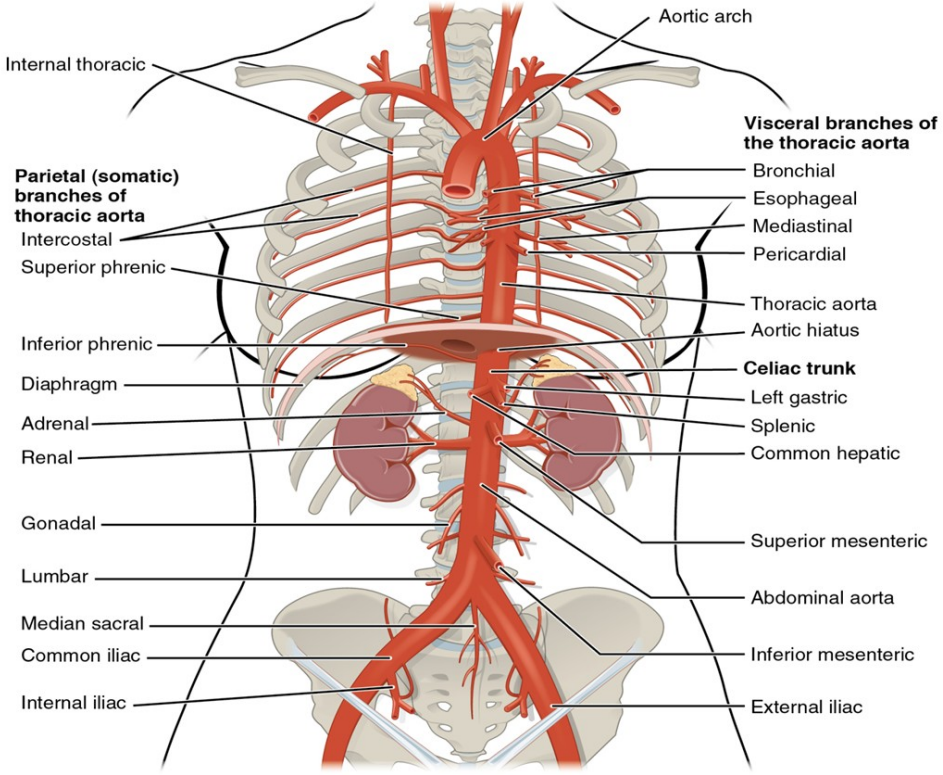
1. Leaves the posterior mediastinum at the level of the T 12th vertebra.
2. By passing behind the diaphragm b/w the crura i.e. behind the median arcuate ligament.



Source: Morton DA, Foreman KB, Albertine KH: *The Big Picture: Gross Anatomy:* www.accessmedicine.com
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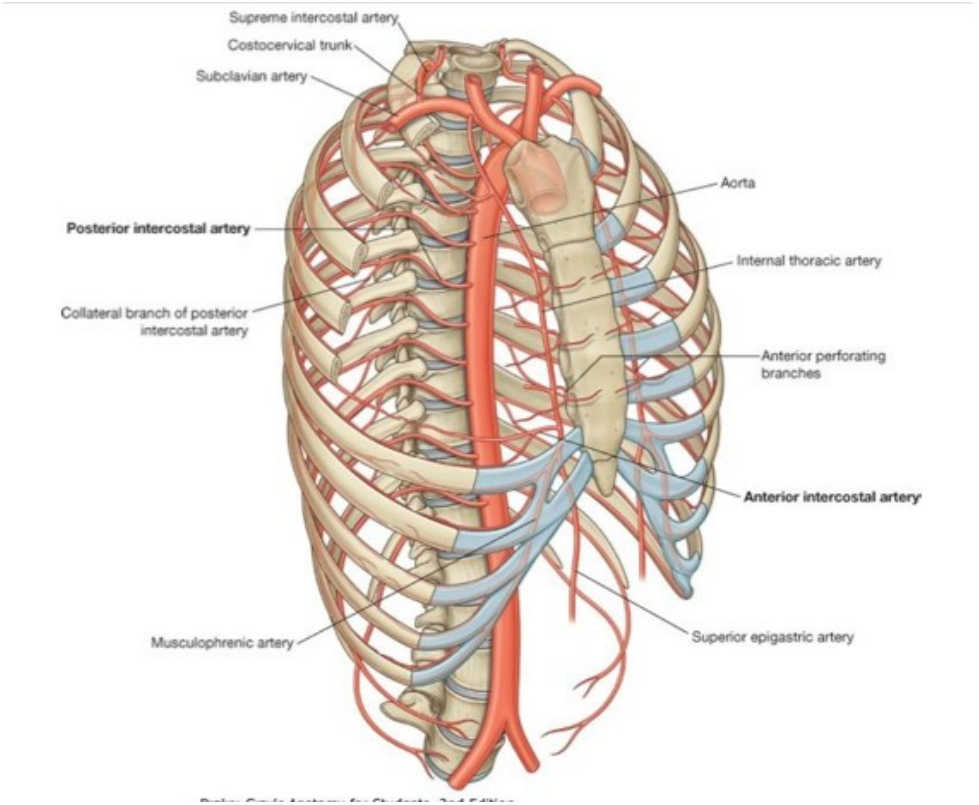
Branches of the descending thoracic aorta

1. Nine pairs of the posterior intercostal arteries.
    1. On each side from 3rd to 11th intercostal spaces.
2. Pair of subcostal arteries.
3. Bronchial arteries.
    1. Two left bronchial arteries.
4. Oesophageal vessels.
    1. Supplying middle 1/3 of the oesophagus.
5. Few pericardial branches.
    1. Posterior surface of the pericardium.



Branches of the descending thoracic aorta

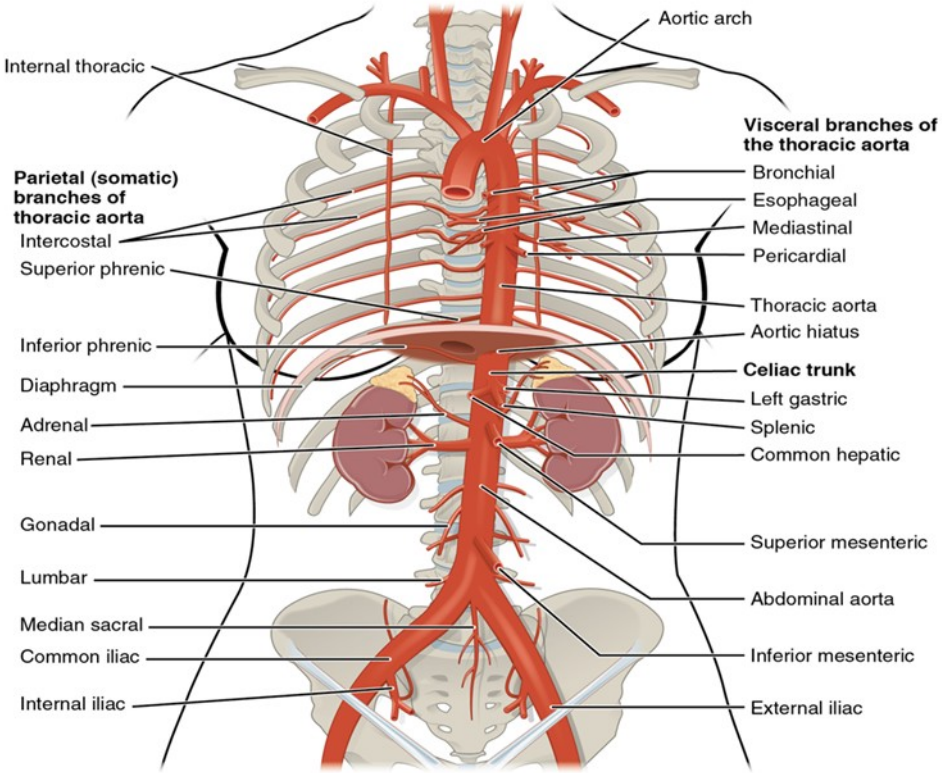
- Mediastinal branches to
  - Lymph nodes & aerolar tissue of the posterior mediastinum.
- Superior phrenic arteries
  - Anastomose with branches of musculophrenic & pericardiophrenic arteries.



Internal thoracic artery

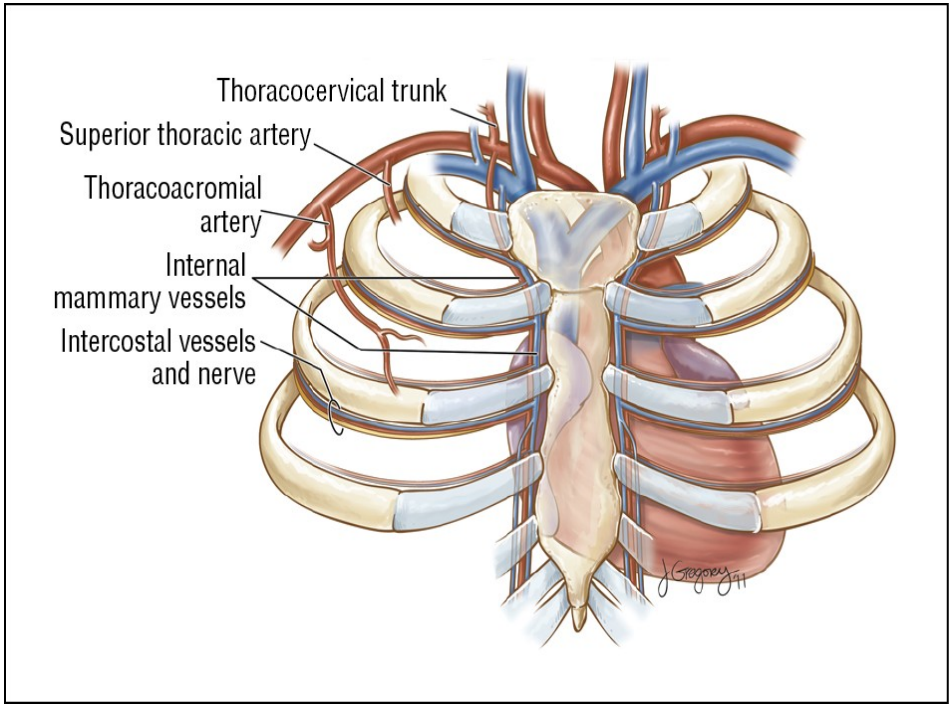
- Origin.

1. Arise from 1st part of the subclavian artery.
2. 2cm above the sternal end of the clavicle.



Course of the internal iliac artery

- Above the 1st costal cartilage.
- It runs downwards & medially behind the
  - Sternal end of the clavicle.
  - Internal jugular vein.
  - Brachiocephalic vein.
  - 1st costal cartilage.
  - phrenic
- It descends in front of the
  - Cervical pluera.



Course of the internal iliac artery

- Below the 1st costal cartilage.
- It runs vertically downwards 1cm lateral to the lateral border of the sternum up to 6th intercostal space
- Terminate by dividing into
  - Superior epigastric &
  - Musculophrenic.
- Artery is accompanied by two venacomitants.
- Chain of the lymph nodes lies along the artery.

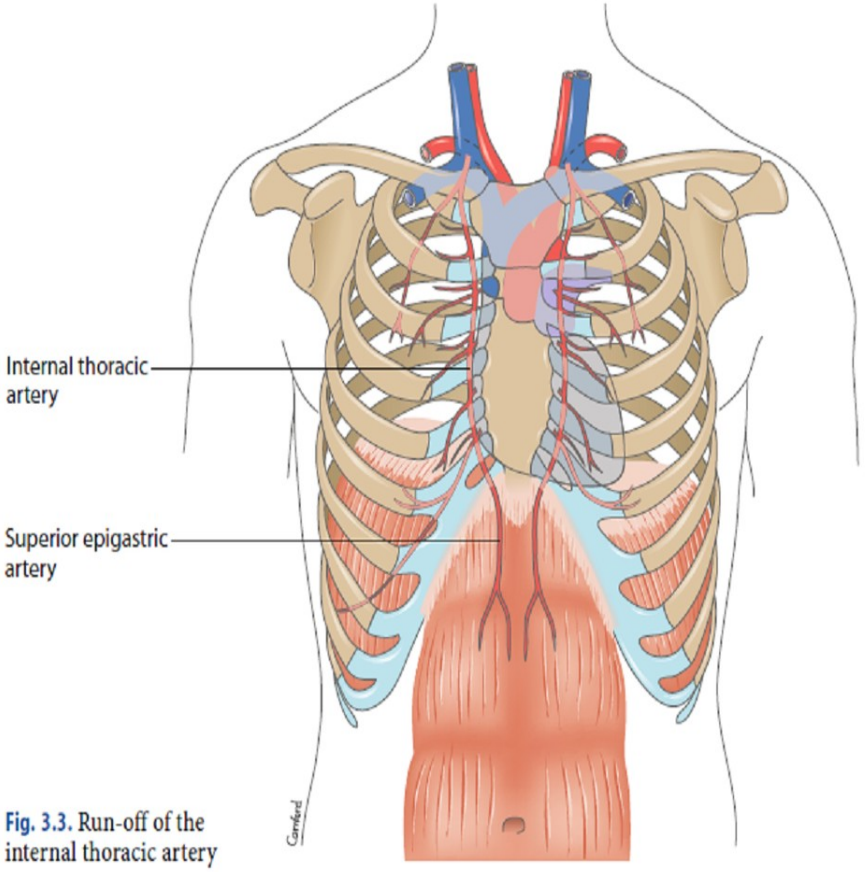
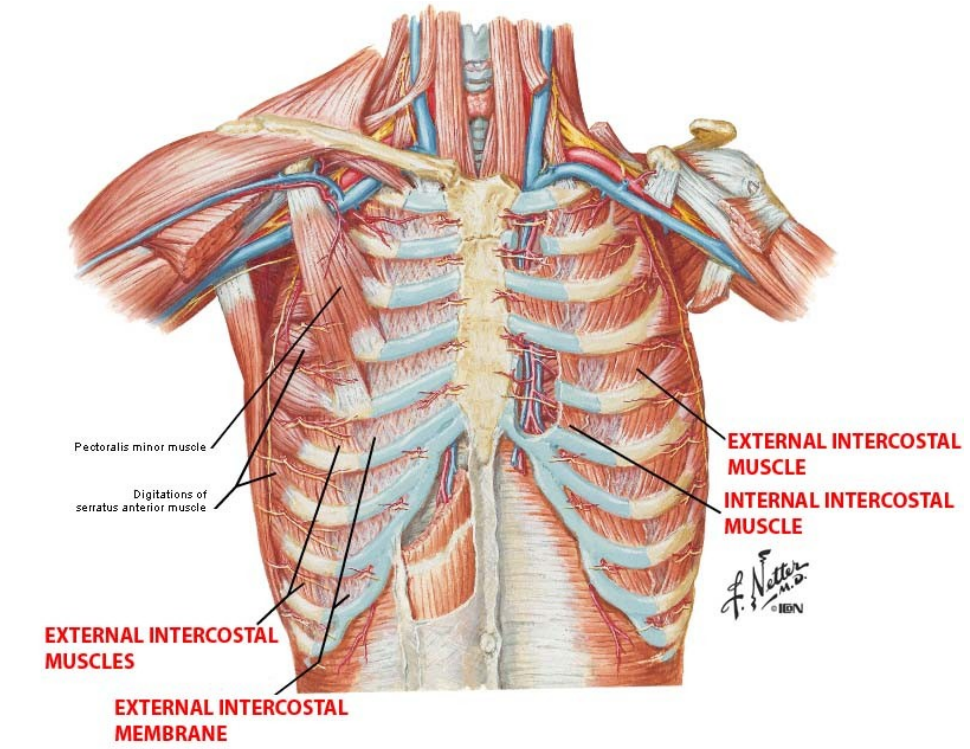


Fig. 3.3. Run-off of the internal thoracic artery

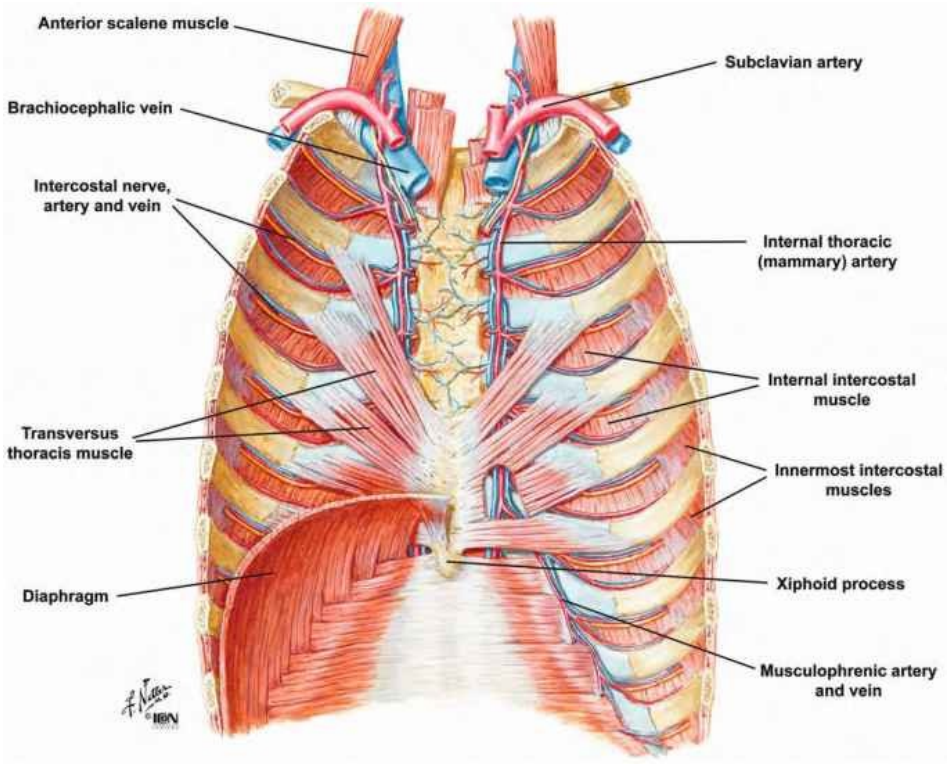
Anterior relations of the internal thoracic artery

1. Pectoralis major muscle.
2. Upper six costal cartilage.
3. External intercostal membrane.
4. Internal intercostal muscles.
5. 1st six intercostal nerves.



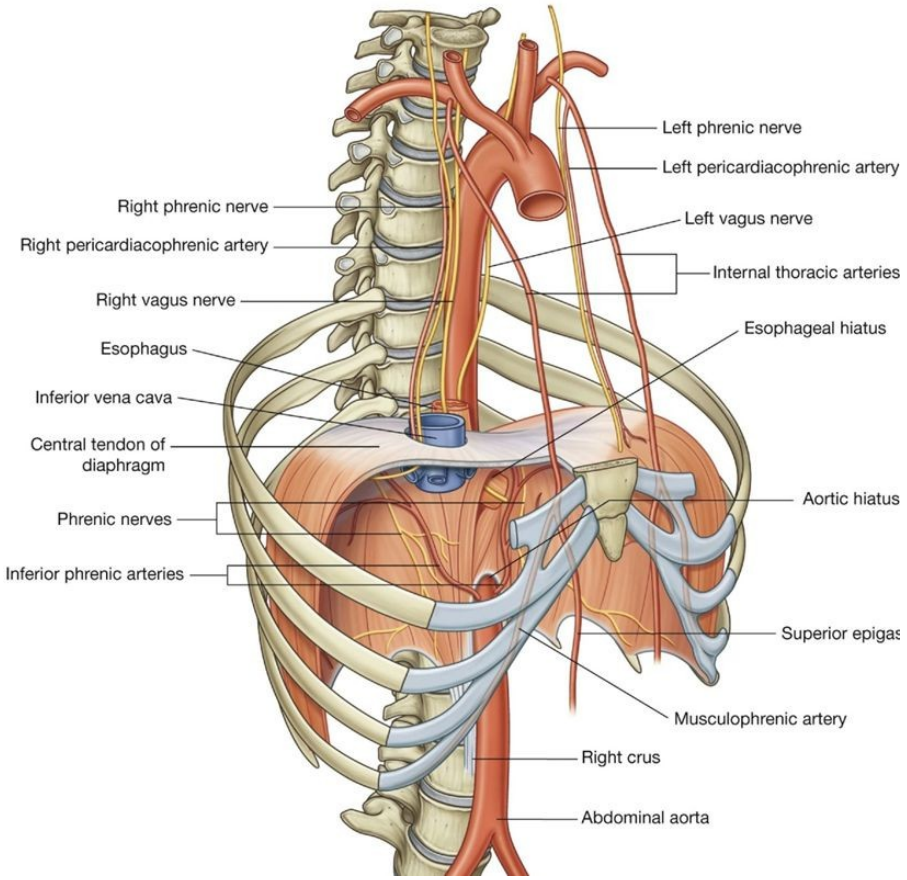
posterior relations of the internal thoracic artery

1. Endothoracic fascia & pleura up to 2nd or 3rd costal cartilage.
2. Below sternocostalis muscle separate the artery from pleura.



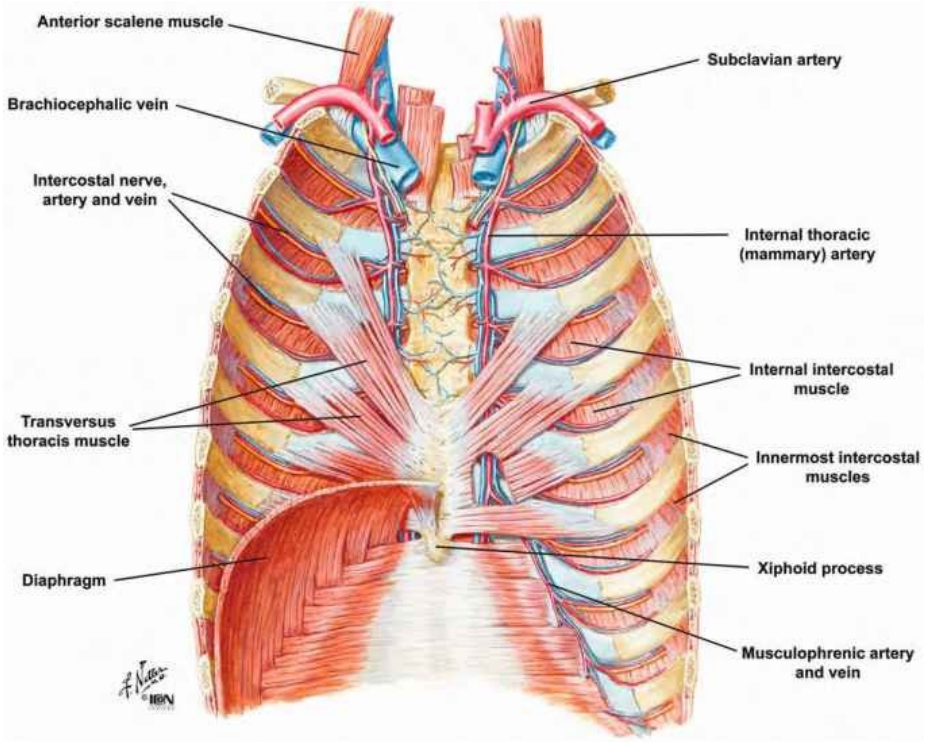
Branches of the internal thoracic artery

1. Pericardiophrenic artery.
    1. Arise at the root of the neck.
    2. Accompanies the phrenic nerve.
    3. Reaches the diaphragm.
    4. Supplies the pleura & pericardium.



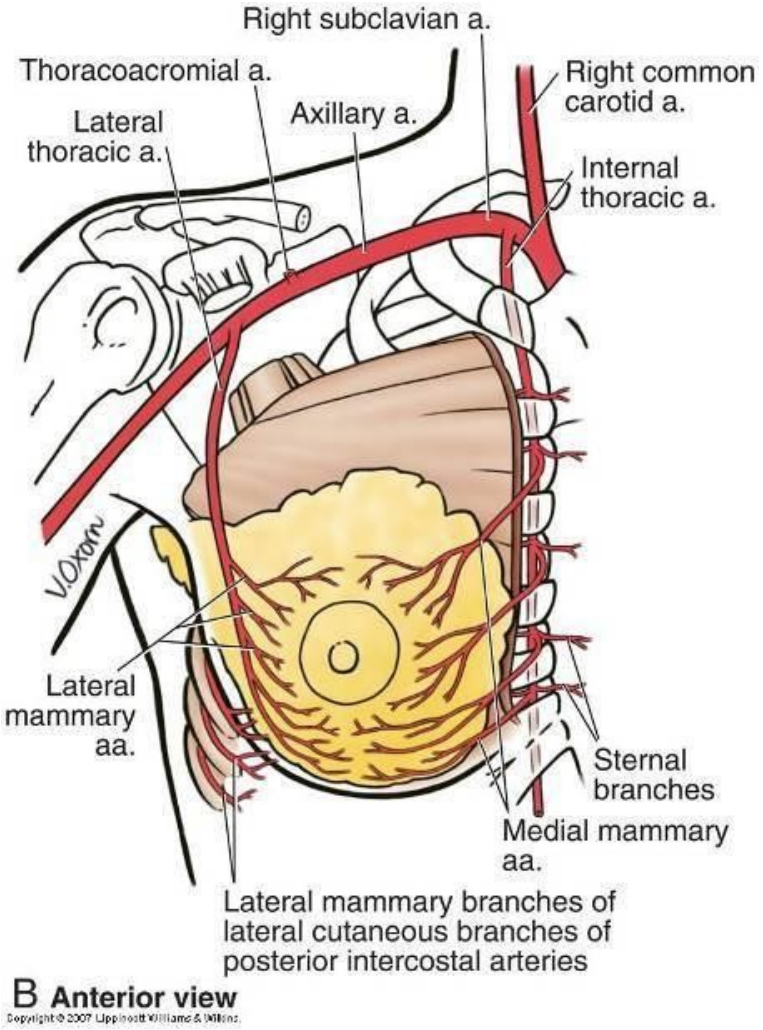
Branches of the internal thoracic artery

- Two anterior intercostal arteries to each of the upper six intercostal spaces.



Branches of the internal thoracic artery

- Mediastinal branches arteries supplies
  - Thymus
  - In front of the pericardium.
  - fat in the mediastinum.
- Perforating arteries accompany the anterior cutaneous nerves.
  - In female perforating arteries in the 2nd, 3rd &4th spaces are large & supply the breast.



Branches of the internal thoracic artery

- Superior epigastric artery.
  - Runs behind the 7th costal cartilage.
  - Enter the rectus sheath by passing b/w the sternal & costal slips of origin of the diaphragm.

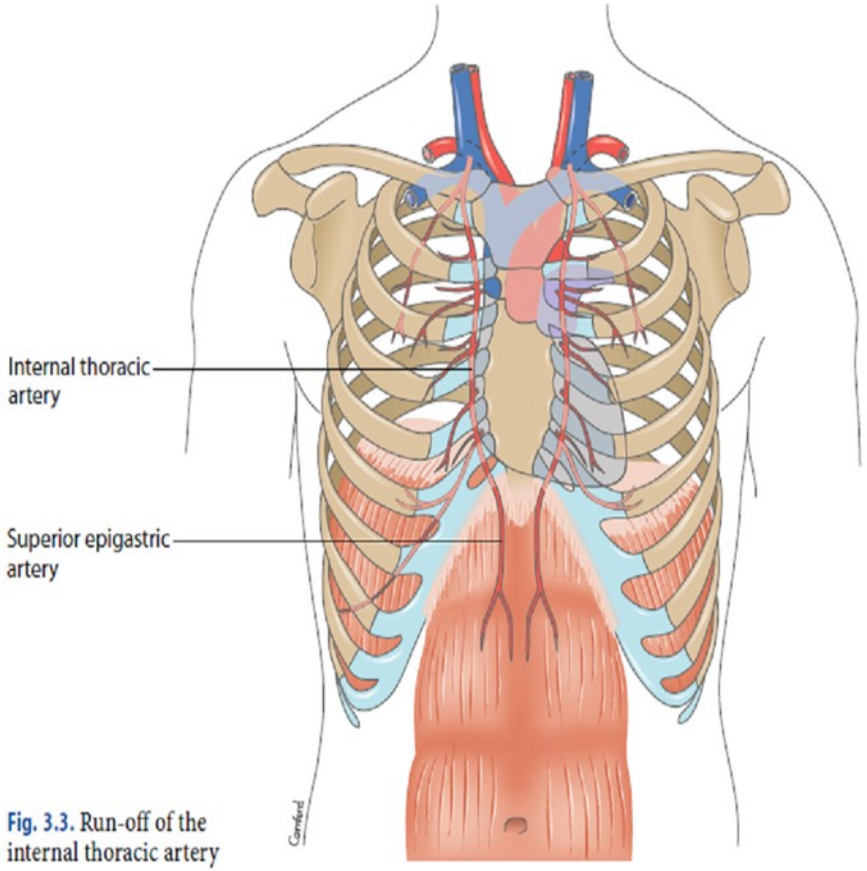
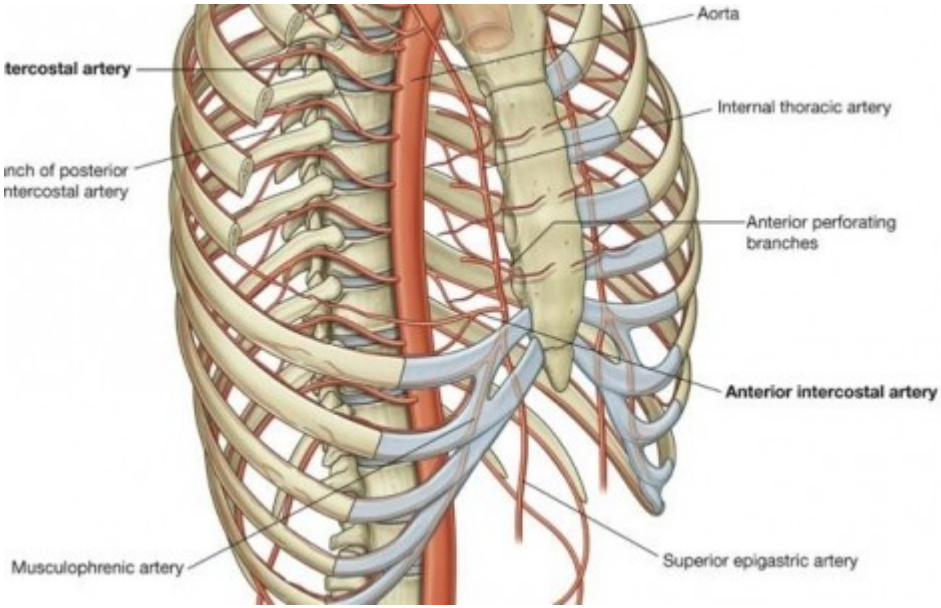


Fig. 3.3. Run-off of the internal thoracic artery

Branches of the internal thoracic artery

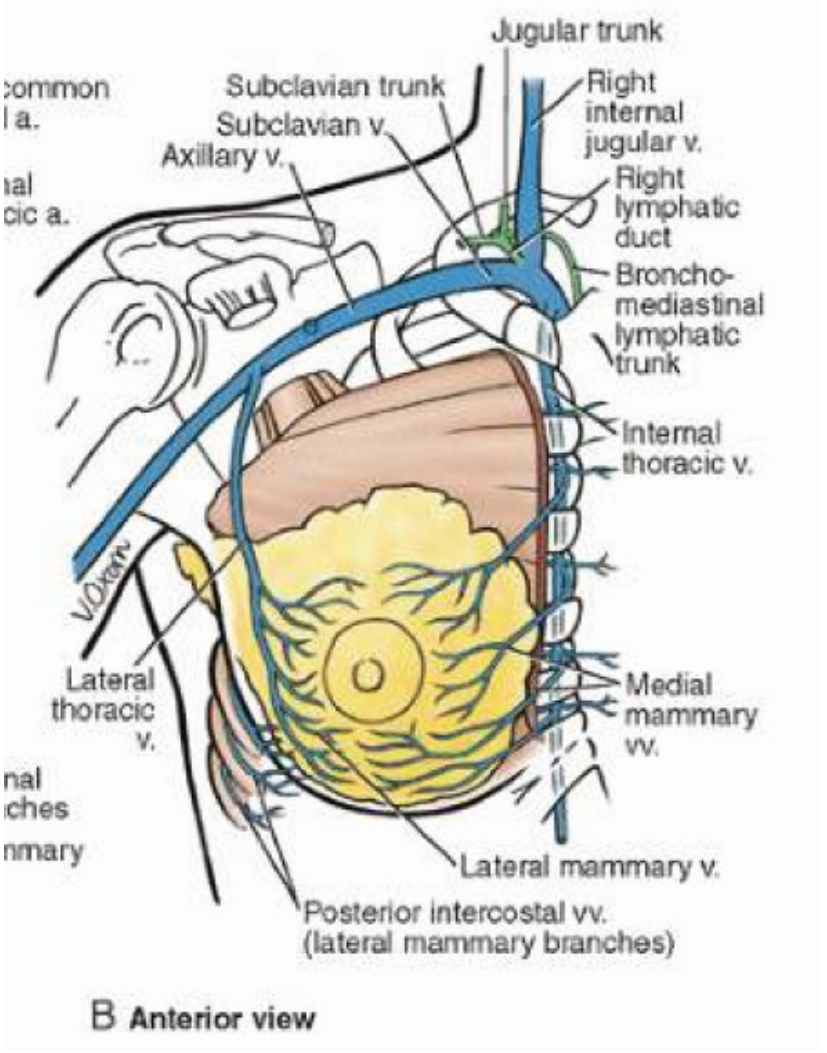
- Musculophrenic artery.

1. Runs downwards & laterally .
2. Behind the 7th , 8th & 9th costal cartilages.
3. Gives two anterior intercostal arteries in each of these spaces.
4. Perforate the diaphragm near 9th costal cartilages
5. Terminates by anastomosing with other arteries on undersurface of the diaphragm.



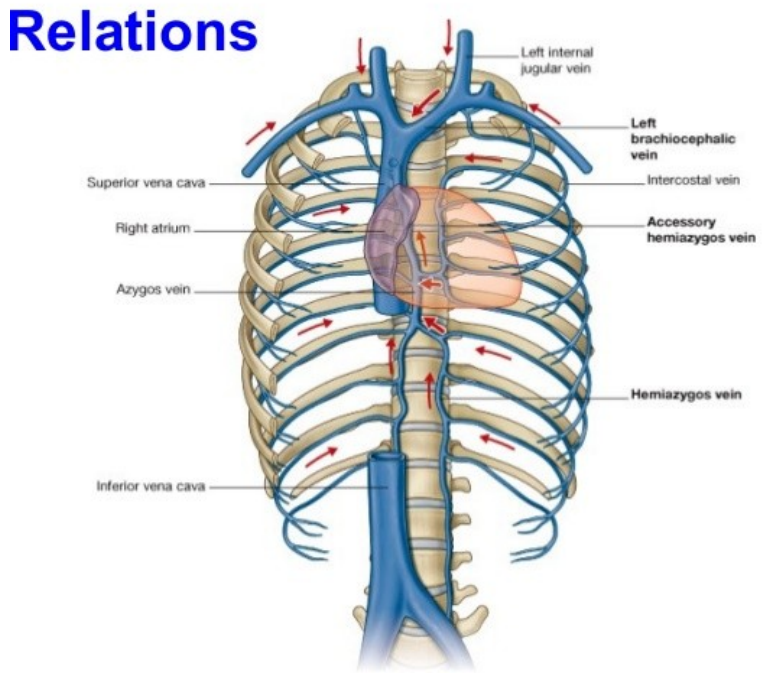
Internal thoracic vein

- Two venacomitantes unite at the level of the 3rd costal cartilage to form the internal thoracic vein.
- Vein runs upwards along the medial side of the artery .
- End in the brachiocephalic vein at the inlet of the thorax.

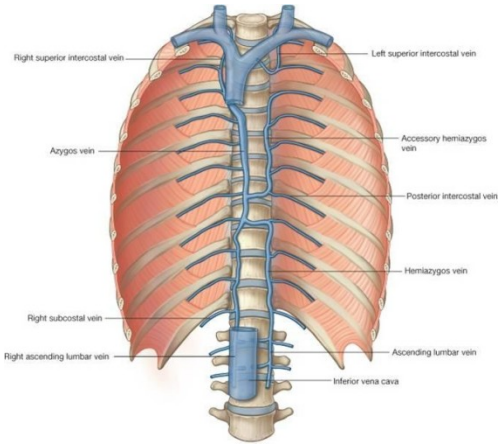


Azygos vein

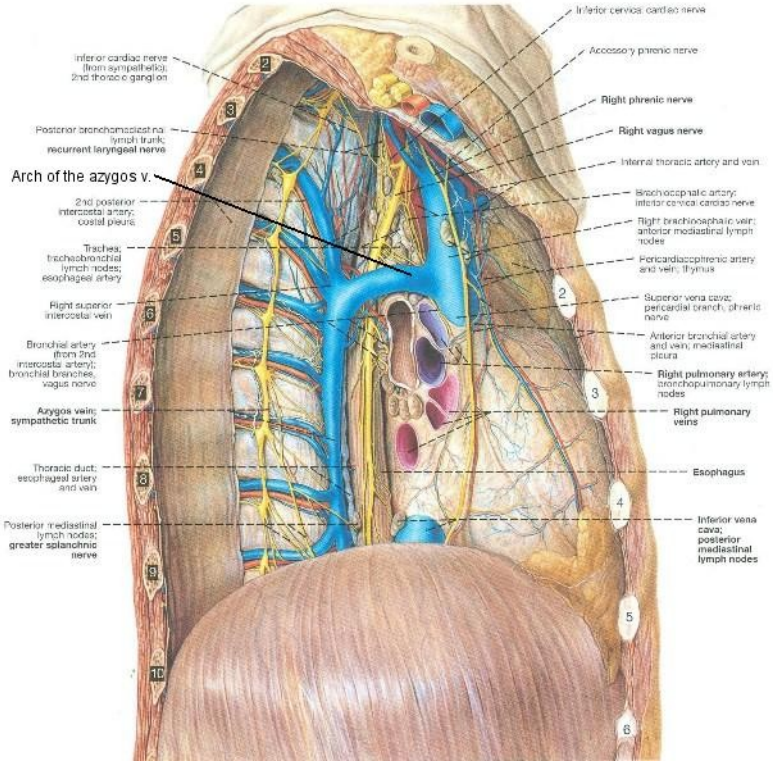
1. Drains the thoracic wall & upper abdomen.
2. Forms an important connecting channel b/w superior & inferior venacava.
3. Occupies posterior abdominal wall & posterior mediastinum.
4. Lower end is connected to the inferior venacava.



Azygos vein

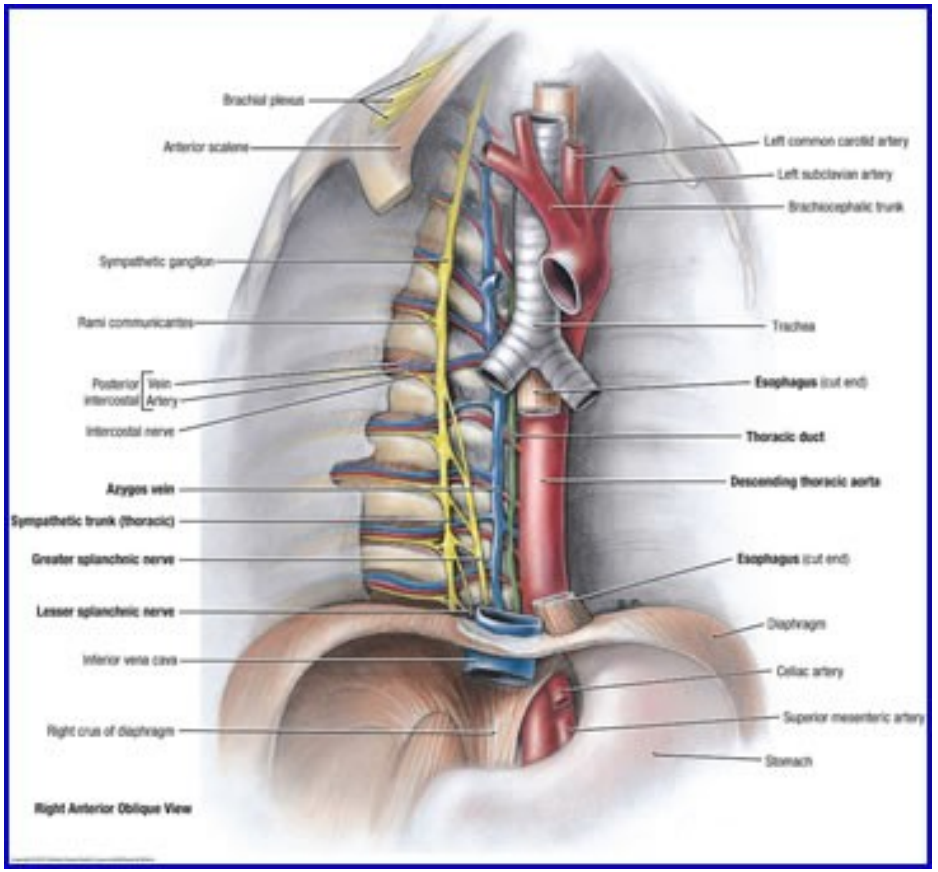


1. Formed by the union of the ascending lumbar vein with subcostal vein on the right side.
2. Enter the thorax by passing through aortic opening of the diaphragm.
3. Ascends up to the fourth thoracic vertebra.
4. Arches over the root of the right lung.
5. Ends by joining the posterior aspect of the superior venacava just before it pierce the pericardium.



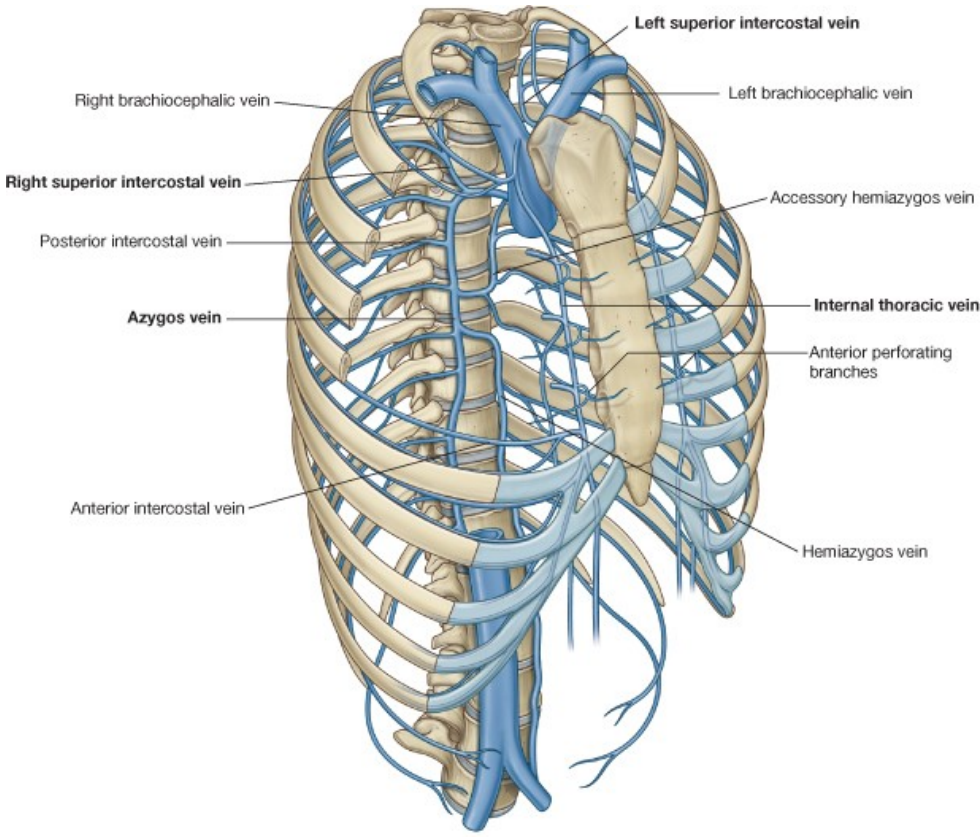
Relation of the azygos vein

1. Anterior.
    1. Oesophagus.
2. Posterior.
    1. Lower eight thoracic vertebrae.
    2. Lower right posterior intercostal arteries.
3. Right
    1. Right lung & pleura.
    2. Greater splanchnic nerve.
4. Left
    1. Thoracic duct & aorta in the lower part
    2. Oesophagus, trachea & vagus in the upper part.



Tributaries of the azygos vein

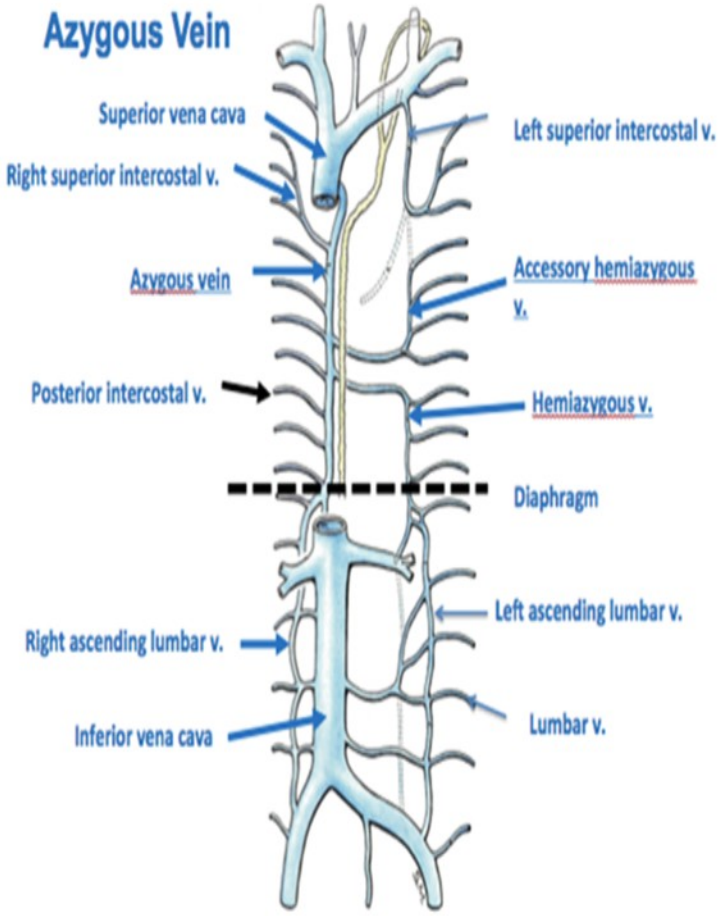
1. Right superior intercostal vein.
    1. Formed by union of the 2nd.3rd& 4th posterior intercostal veins.
2. 5th to 11th right posterior intercostal veins.
3. Hemiazygos vein at T 8th vertebra.
4. Right bronchial vein.
5. Several oesophageal, mediastinal & pericardial veins.



Hemiazygos vein

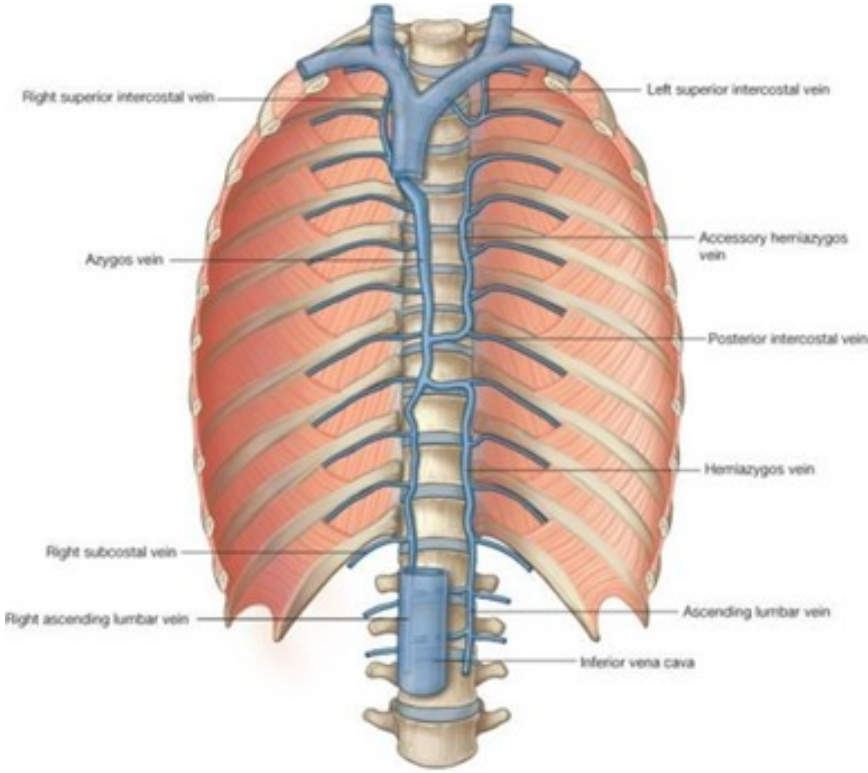
- Two in numbar

1. Superior one is called hemiazygos vein.
2. Formed by union of the left ascending lumbar & left subcostal vein.



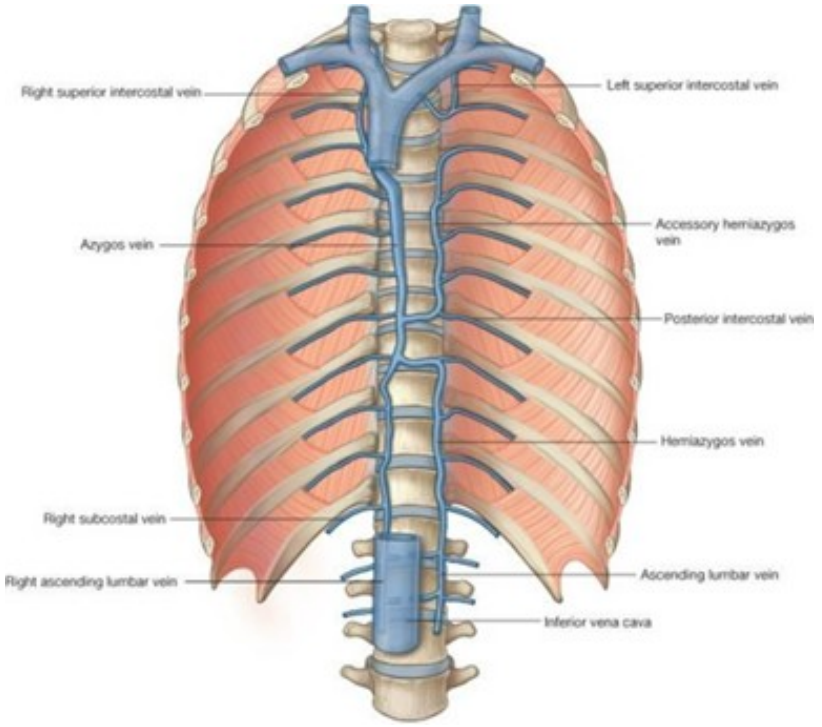
Hemiazygos vein

1. Course.
    1. Pierce the left crus of the diaphragm.
    2. Ascends on the left side of the vertebra.
    3. Overlapped by the aorta.
2. Termination.
    1. At T 8th vertebra it turns to the right side .
    2. Passes behind the oesophagus & thoracic duct .
    3. Join the azygos vein.



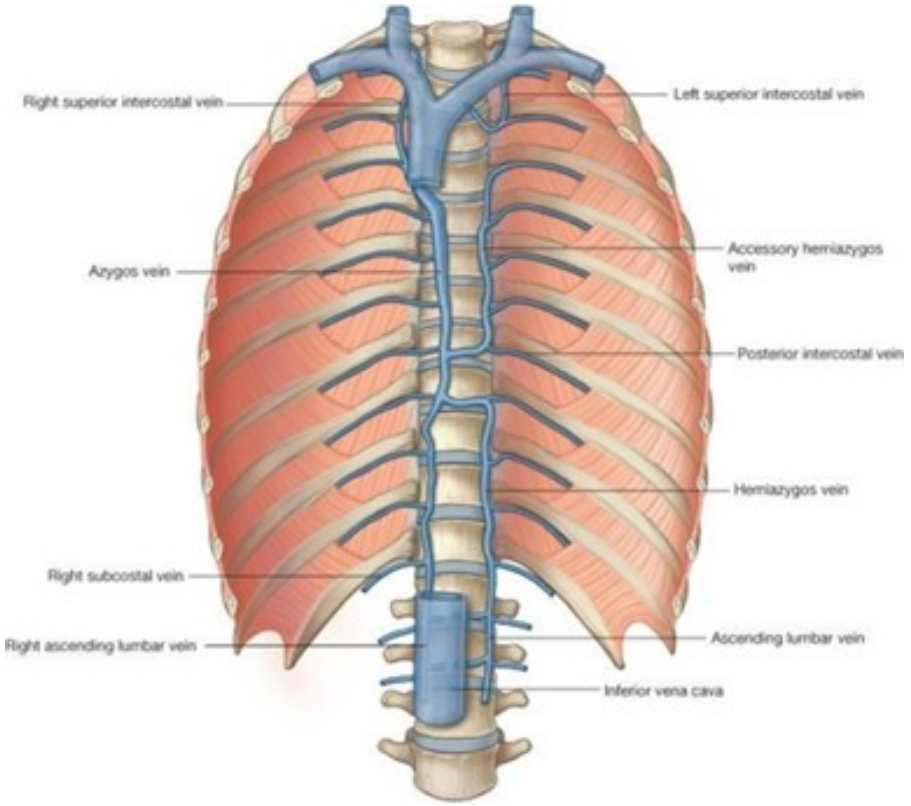
Tributaries of the hemiazygos vein

1. Left ascending lumbar.
2. Left subcostal
3. 9-11 posterior intercostal veins.



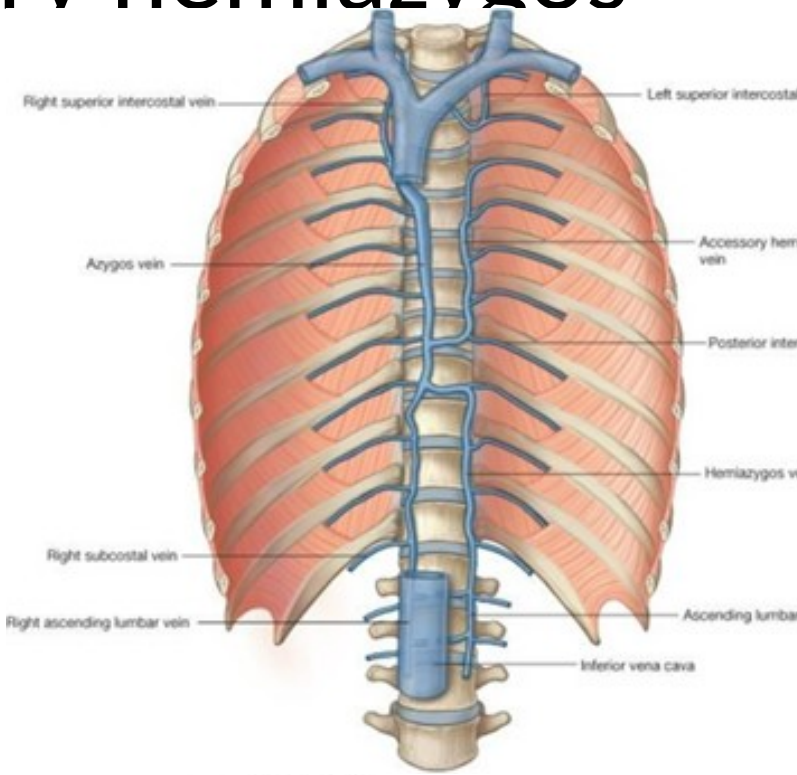
Accessory hemiazygos vein

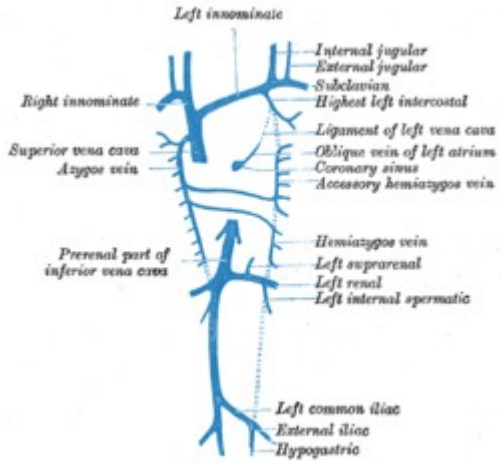
1. Also called the superior hemiazygos vein
2. Formed by confluence of the left middle intercostal veins.
3. Begins at the medial end of the 4th or 5th left intercostal space.
4. Descends on the left side of the vertebral collumn.
5. Turns to the left
6. Passess behind the aorta & thoracic duct.
7. Join the azygos vein.



Tributaries of accessory hemiazygos vein

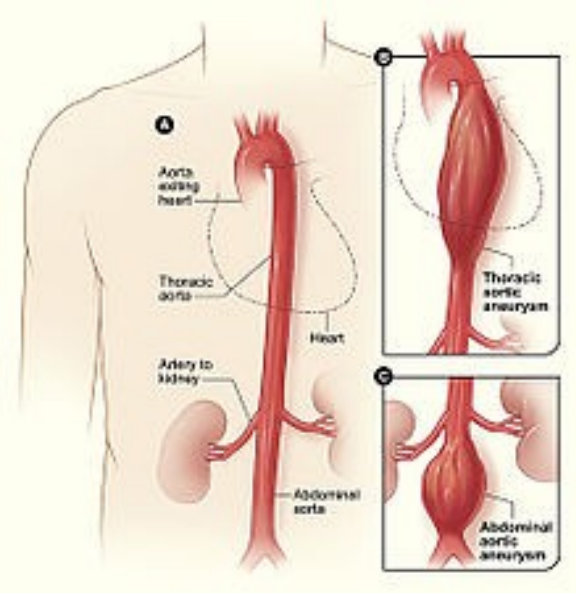
1. 5th to 8th left posterior intercostal veins.
2. Sometime left bronchial veins.





Clinical anatomy

1. Aortic aneurysm.
2. Aortic stenosis.
3. Azygos system provide collateral pathway of venous drainage in superior or inferior venous obstruction.



THANKYOU